

MATH 127 Calculus for the Sciences

Lecture 30

Today's lecture

Last time

Global maxima and minima

This time

Course note coverage Section 4.1.2, 4.2.1

Optimization

linearization

Optimization problems

You have learnt how to find global maxima and global minima.

To translate that to real word problem, you are trying to optimize/find the best thing that does a thing.

Example If you drink x amount of coffee, you will be awake at a level of y . But if you drink too much, you will die and your level of awakesness is negative. Suppose

$$y = -(x - 1)^2 + 1$$

Find the amount of coffee you should drink to get maximal awakesness.

This examples asks you to “optimize” coffee drinking, or in mathematical terms, to find a global maximum of y .

Optimization problems

Steps to solve real world optimization problem

1. Create a math model for the problem, so that the problem becomes looking for maximum/minimum of some function.
2. If the function is defined on a closed interval, then use closed interval method.
3. If the function is defined on the entire real line, what do we do?

Example

Question You are preparing a cylindrical jar that could fit 10 balls, each with a different number, numbers 1 to 10. The jar must have a radius of at least 1cm for the balls to fit, and should be at least 2cm tall. It must have a volume of exactly 100cm^3 . Next to avoid tempering with the psychic abilities of your audience, you need to glue black paper around the side of the jar. But black paper are extremely expensive, so you want to use as little as possible. How tall and how wide should the jar be?

look for minimum

1. The volume, in terms of radius and height, is:

$$\text{Volume} = \text{Area of disc} \times \text{height} = \pi r^2 \cdot h = 100$$

2. The height, in terms of radius, is:

$$h = \frac{100}{\pi r^2}$$

3. The area of the black paper, in terms of radius, is

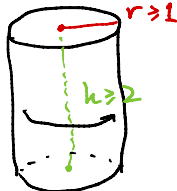
$$\text{Area of paper} = 2\pi r \cdot h = 2\pi r \cdot \frac{100}{\pi r^2} = \frac{200}{r}$$

4. The valid domain for r is

We have $r \geq 1$

$$h \geq 2, \quad \frac{100}{\pi r^2} \geq 2, \quad \text{so } r^2 \leq \frac{50}{\pi}, \quad r \leq \sqrt{\frac{50}{\pi}}$$

so domain for r is $[1, \sqrt{\frac{50}{\pi}}]$.



Volume = 100

Example

Question We have reduced the problem to finding a global minimum of the function

$$\text{Area}(r) = 2h\pi r = 2 \cdot \frac{100}{\pi r^2} \cdot r = \frac{200}{r}$$

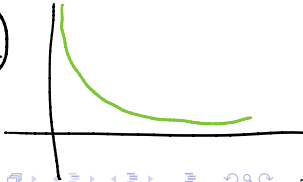
on the interval $\left[1, \sqrt{\frac{50}{\pi}}\right]$.

There are two ways you can proceed from here

1. Use **closed interval method**, make sure you remember to mention the interval is indeed **closed**, and the function is **continuous**.
2. Look at the graph of $\frac{200}{r}$. On the positive side, it is always

decreasing, so it reaches minimum at $\sqrt{\frac{50}{\pi}}$.

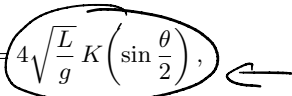
$$h = \frac{100}{\pi r^2} = \frac{100}{\pi \cdot \left(\sqrt{\frac{50}{\pi}}\right)^2} = 2$$



Linearization

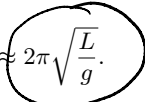
Some times a physics formula can be really bad:

If we try to obtain the exact period T for a pendulum released from an initial angle θ , we find

$$T(\theta) = 4\sqrt{\frac{L}{g}} K\left(\sin \frac{\theta}{2}\right),$$


where K is an elliptic integral of the first kind.

That sounds nice, but physicists are scared of math, so they take a less accurate, but simplified version

$$T \approx 2\pi\sqrt{\frac{L}{g}}.$$


So they say every pendulum of the same length has the same period, but that is in fact not true. Scientists lie to you but math does not!

This simplified version is called **linearization**.

Linearization

Definition For a function $f(x)$, its **linearization** at a is

$$L_a(x) = f(a) + f'(a)(x - a)$$

Definition For a function $f(x)$, its **linear approximation** at a is

$$f(x) \approx L_a(x) = f(a) + f'(a)(x - a)$$

Haven't we already seen this? This is also called the

tangent line approximation.

We are not learning anything new! We are doubly learning something old!

Example

Example Suppose occult experts claim that the *spookiness* S of a room depends on the *number of haunted souls* h according to the formula

$$S(h) = \sqrt{1 + 3h^2} \sin h.$$

Question: Find the linearization of $S(h)$ at $h = 0$ and approximate the spookiness given that there are 0.03 haunted souls.

tangent line
tangent line approximation

$$1. \quad S'(h) = \frac{1}{2} \cdot \frac{1}{\sqrt{1+3h^2}} \cdot 6h \cdot \sin h + \underbrace{\sqrt{1+3h^2}} \cdot \cos h$$

$$S'(0) = 0 + 1 = 1$$

$$\begin{aligned} L_0(h) &= S(0) + S'(0)(h+0) \\ &= 0 + 1(h+0) \\ &= h \end{aligned}$$

$$2. \quad S(0.03) \approx L_0(0.03) = 0.03$$